

Unit 9

Systems Thinking

3 Slibraries · 15 framework sheets

SLIBRARY 25 Iceberg Model

5 sheets

- | | |
|----------------------|------------------------|
| 1 Iceberg · 4 Levels | 4 Systems Boundaries |
| 2 Events → Patterns | 5 Double-Loop Learning |
| 3 Surface vs Deep | |

SLIBRARY 26 Meadows' Leverage Points

5 sheets

- | | |
|------------------------|-----------------------|
| 1 12 Leverage Points | 4 Goals of the System |
| 2 Parameters & Buffers | 5 Paradigm Shift |
| 3 Feedback Loops | |

SLIBRARY 27 Archetypes & Theory U

5 sheets

- | | |
|--------------------------|--------------|
| 1 Systems Archetypes (8) | 4 Theory U |
| 2 Shifting the Burden | 5 Presencing |
| 3 Tragedy of the Commons | |

Iceberg · 4 Levels

SLIBRARY 25 · SHEET 1 / 5

Events · Patterns · Structures · Mental Models

PROFESSIONAL DEFINITION

The Iceberg Model is a four-level diagnostic frame for systems analysis: visible *Events* (what happens), repeating *Patterns* (what has happened over time), underlying *Structures* (relationships and incentives that produce the patterns), and the deepest layer of *Mental Models* (assumptions and beliefs that shape the structures). Most managerial action targets events; deep leverage lives in structures and mental models, where most leaders never look (Senge, 1990; Meadows, 2008).

WHO DEVELOPED IT

The Iceberg Model emerged from Peter Senge's *The Fifth Discipline* (1990) at MIT and was elaborated by Donella Meadows in *Thinking in Systems* (2008). Roots trace to Jay Forrester's (1961) system-dynamics tradition at MIT and the broader cybernetics literature (Wiener, 1948; Bateson, 1972).

WHY IT WAS DEVELOPED

Managers responding to crises optimised event-level fixes — patches that produced the same crisis again. Senge and Meadows argued that recurring problems traced to deeper structural and mental-model causes; only fixes at those depths produced durable change.

HOW IT IS USED

Walk the four levels in sequence. Events: what happened? Patterns: has this happened before? Structures: what relationships produce the pattern? Mental Models: what assumptions hold the structures in place? Intervention depth determines durability.

WHEN IT IS USED

Use in recurring-crisis diagnosis, in change-management design, in policy analysis, in organisational-development interventions, and as the opening lens of any systems-thinking workshop.

BY WHOM IT IS USED

Organisational-development practitioners, public-policy analysts, change-management leaders, transformation officers, and senior leaders facing recurring problems despite repeated event-level fixes.

FIGURE 1 · FOUR DEPTH LEVELS

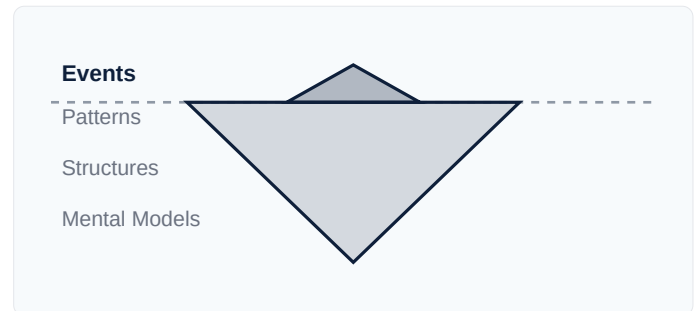


Figure 1. Most action targets events; durable change lives at the structures and mental-models levels.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Where Forrester's (1961) system dynamics required mathematical modelling and Wiener's (1948) cybernetics was technical, Senge's contribution is the four-level diagnostic that any leader can apply. Building on Argyris and Schön's (1974) double-loop learning, the Iceberg gives systems thinking a teachable depth-of-intervention frame.

RELATED FRAMEWORKS IN THIS COURSE

Events → Patterns · Slibrary 25

Surface vs Deep · Slibrary 25

Double-Loop Learning · Slibrary 25

12 Leverage Points · Slibrary 26

SOURCES (HARVARD REFERENCING STYLE)

Senge, P.M. (1990) *The Fifth Discipline: The Art and Practice of the Learning Organization*. New York: Doubleday.
 Meadows, D.H. (2008) *Thinking in Systems: A Primer*. White River Junction: Chelsea Green.
 Forrester, J.W. (1961) *Industrial Dynamics*. Cambridge, MA: MIT Press.
 Argyris, C. and Schön, D.A. (1974) *Theory in Practice*. San Francisco: Jossey-Bass.

Events → Patterns

SLIBRARY 25 · SHEET 2 / 5

One event is anecdote; repeated events are pattern

PROFESSIONAL DEFINITION

Events → Patterns is the diagnostic move from level 1 to level 2 of the Iceberg: tracking a recurring event over time to surface its pattern. A single late delivery is an event; weekly late deliveries are a pattern. The shift from event-thinking to pattern-thinking is the first analytical step in systems diagnosis — it forces the question "why does this keep happening?" rather than "how do we fix this one?". Most chronic problems become tractable only at the pattern level (Senge, 1990; Meadows, 2008).

WHO DEVELOPED IT

Articulated by Peter Senge in *The Fifth Discipline* (1990) as the second level of the Iceberg, drawing on Forrester's (1961) system dynamics. Donella Meadows operationalised the analytical move in workshop practice through her work at MIT and Dartmouth (Meadows, 2008).

WHY IT WAS DEVELOPED

Event-thinking trapped managers in repeated firefighting. Pattern-thinking lifted analysis to the time-series view where structural causes became visible. The move from one to the other is the first cognitive shift required for systems diagnosis.

HOW IT IS USED

Whenever a notable event occurs, ask: has this happened before? Map occurrences over time. Look for cyclicity, escalation, or correlation with other events. The pattern-level view changes which questions are askable.

WHEN IT IS USED

Use whenever a problem recurs, in incident-response post-mortems, in chronic-issue diagnosis, in board-level performance reviews, and as the opening move of any systems analysis.

BY WHOM IT IS USED

Organisational-development practitioners, change-management leaders, operations directors, public-health analysts, and any leader facing recurring issues.

FIGURE 1 · TIME-SERIES VIEW

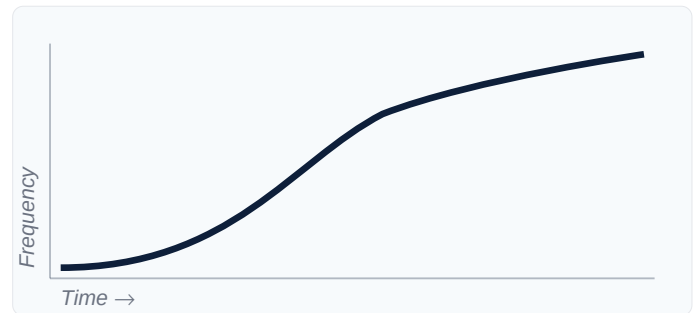


Figure 1. The pattern view changes which questions are askable; structural causes appear only at this level.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Where Forrester's (1961) system dynamics modelled patterns mathematically, Senge's contribution is the cognitive-shift framing that makes pattern-thinking accessible without quantitative modelling. Building on Argyris's (1974) reflective-practice tradition, it gave practitioners the first analytical move of systems diagnosis.

RELATED FRAMEWORKS IN THIS COURSE

Iceberg · 4 Levels · Slibrary 25

Surface vs Deep · Slibrary 25

Feedback Loops · Slibrary 26

Systems Boundaries · Slibrary 25

SOURCES (HARVARD REFERENCING STYLE)

- Senge, P.M. (1990) *The Fifth Discipline*. New York: Doubleday.
 Meadows, D.H. (2008) *Thinking in Systems*. White River Junction: Chelsea Green.
 Forrester, J.W. (1961) *Industrial Dynamics*. Cambridge, MA: MIT Press.
 Sterman, J.D. (2000) *Business Dynamics*. New York: McGraw-Hill.

Surface vs Deep

SLIBRARY 25 · SHEET 3 / 5

Symptomatic fixes vs root-cause interventions — different depths, different durabilities

PROFESSIONAL DEFINITION

Surface vs Deep is the structural distinction between symptomatic fixes (which address visible events) and root-cause interventions (which address underlying structures or mental models). Surface fixes are fast, cheap and politically easy; deep fixes are slow, expensive and politically hard. The asymmetry produces a systematic bias toward surface fixes that produce the same problem repeatedly — the deepest cost of which is missed root-cause learning (Senge, 1990; Meadows, 2008; Goldratt, 1984).

WHO DEVELOPED IT

Articulated across the systems-thinking literature: Senge's (1990) Iceberg-depth framing, Meadows's (2008) leverage-point analysis, and Eli Goldratt's (1984) Theory of Constraints — all converge on the surface-vs-deep distinction. The tradition traces to Forrester's (1961) original system-dynamics work.

WHY IT WAS DEVELOPED

Organisations defaulted to surface fixes because they were politically and economically rewarded; deep fixes were punished by their slow payoff. Naming the asymmetry made it discussable — and made the structural bias toward surface fixes a deliberate design choice rather than an unconsidered default.

HOW IT IS USED

For each problem, identify candidate fixes at multiple depths. Calculate true cost (including recurrence cost of surface fixes). Surface fixes are appropriate for genuine one-offs; deep fixes are necessary for recurring patterns. Match depth to problem type.

WHEN IT IS USED

Use in transformation programme design, in operations-improvement decisions, in policy-intervention design, and as a check against teams' default toward surface fixes.

BY WHOM IT IS USED

Senior leaders sponsoring transformation, operations directors, public-policy analysts, organisational-development consultants, and change-management practitioners.

FIGURE 1 · DEPTH × DURABILITY

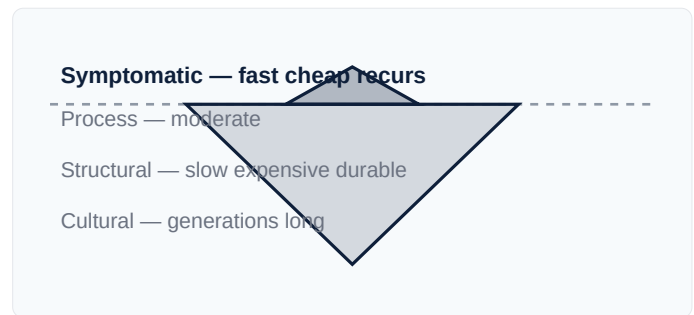


Figure 1. Surface fixes are politically rewarded but produce recurrence; depth correlates with durability.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Where Goldratt's (1984) Theory of Constraints emphasised bottleneck-as-root-cause and Toyota's (Ohno, 1988) 5 Whys drove root-cause analysis, the surface-vs-deep distinction's contribution is the explicit cost-asymmetry framing. Building on Forrester's (1961) system-dynamics tradition, it makes intervention depth a deliberate strategic decision.

RELATED FRAMEWORKS IN THIS COURSE

[Iceberg · 4 Levels · Slibrary 25](#)

[Events → Patterns · Slibrary 25](#)

[12 Leverage Points · Slibrary 26](#)

[Double-Loop Learning · Slibrary 25](#)

SOURCES (HARVARD REFERENCING STYLE)

- Senge, P.M. (1990) *The Fifth Discipline*. New York: Doubleday.
 Meadows, D.H. (2008) *Thinking in Systems*. White River Junction: Chelsea Green.
 Goldratt, E.M. (1984) *The Goal: A Process of Ongoing Improvement*. Great Barrington: North River Press.
 Ohno, T. (1988) *Toyota Production System*. Cambridge, MA: Productivity Press.

Systems Boundaries

SLIBRARY 25 · SHEET 4 / 5

Where you draw the line determines what you can fix

PROFESSIONAL DEFINITION

Systems Boundaries is the analytical decision of where to draw the system's edge — what counts as inside (subject to intervention) and outside (taken as given). The choice determines which causes are visible and which fixes are conceivable. Drawing too narrowly excludes root causes; drawing too broadly produces analysis paralysis. Boundary-setting is the system designer's most consequential and least examined choice (Meadows, 2008; Checkland, 1981).

WHO DEVELOPED IT

Peter Checkland's *Systems Thinking, Systems Practice* (1981) at Lancaster University introduced soft-systems methodology with explicit boundary-setting. Donella Meadows (2008) operationalised boundary-choice in workshop practice. Roots in Bertalanffy's (1968) general-systems theory.

WHY IT WAS DEVELOPED

Practitioners assumed system boundaries were objectively given by the problem. Checkland and Meadows argued boundaries were analytical choices that shaped what could be seen and fixed — making boundary-setting a deliberate, contestable, sometimes-iterated decision rather than an inheritance.

HOW IT IS USED

Identify candidate boundaries (operational, organisational, ecological, temporal). For each, ask: what is included? What is excluded as given? What causes does the boundary make invisible? Iterate until the boundary captures the relevant causes without paralysing the analysis.

WHEN IT IS USED

Use at the start of any systems analysis, in policy-design where regulatory scope is contested, in M&A; integration where organisational boundaries are being redrawn, and as a diagnostic when interventions consistently fail to address root causes.

BY WHOM IT IS USED

Systems-thinking practitioners, public-policy analysts, organisational-development consultants, M&A; integration leaders, and senior leaders sponsoring transformation programmes.

FIGURE 1 · BOUNDARY CHOICE

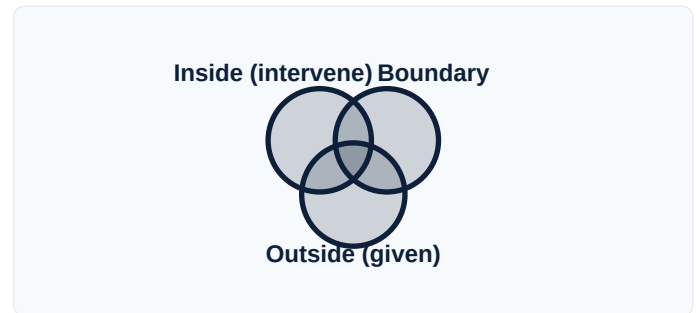


Figure 1. The boundary is an analytical choice; where you draw it determines what you can fix.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Where Bertalanffy's (1968) general-systems theory treated systems as objectively bounded, Checkland's contribution is the soft-systems framing where boundary-choice is constructive and contestable. Building on Vickers's (1968) appreciation-systems tradition, it gives systems practitioners a deliberate design move.

RELATED FRAMEWORKS IN THIS COURSE

Iceberg · 4 Levels · Slibrary 25

Feedback Loops · Slibrary 26

Goals of the System · Slibrary 26

Theory U · Slibrary 27

SOURCES (HARVARD REFERENCING STYLE)

Checkland, P. (1981) *Systems Thinking, Systems Practice*. Chichester: Wiley.
 Meadows, D.H. (2008) *Thinking in Systems*. White River Junction: Chelsea Green.
 von Bertalanffy, L. (1968) *General System Theory*. New York: Braziller.
 Vickers, G. (1968) *Value Systems and Social Process*. London: Tavistock.

Double-Loop Learning

SLIBRARY 25 · SHEET 5 / 5

Argyris — fix the error vs fix the rule that produced the error

PROFESSIONAL DEFINITION

Chris Argyris's Double-Loop Learning distinguishes two levels of organisational learning: *single-loop* learning corrects errors against existing goals and rules; *double-loop* learning revises the goals and rules themselves when single-loop correction fails repeatedly. Single-loop is the default; double-loop is rare because it requires examining and challenging the assumptions that govern action. Without double-loop capability, organisations repeat the same structural errors at higher fidelity (Argyris and Schön, 1974, 1996).

WHO DEVELOPED IT

Chris Argyris and Donald Schön at Harvard, in their 1974 book *Theory in Practice* and elaborated through Argyris's subsequent *Reasons and Rationalizations* (2004) and the Schön-co-authored *Organizational Learning II* (1996). Roots in Lewin's (1947) action-research tradition.

WHY IT WAS DEVELOPED

Organisations facing repeating problems intensified single-loop fixes (more rules, more controls) without examining the rules themselves. Argyris's diagnosis: the rules were the problem, and only double-loop learning could surface and revise them. The distinction made the problem visible and addressable.

HOW IT IS USED

When single-loop fixes fail repeatedly, switch to double-loop: surface the governing variables (goals, rules, mental models) producing the failure pattern. Make assumptions discussable. Revise governing variables. Test new behaviour against revised assumptions.

WHEN IT IS USED

Use when single-loop fixes have failed repeatedly, in transformation-programme design, in organisational-culture change, and as a diagnostic for chronic structural problems.

BY WHOM IT IS USED

Organisational-development consultants, change-management leaders, executive coaches, transformation officers, and senior leaders sponsoring deep cultural change.

FIGURE 1 · TWO LOOPS

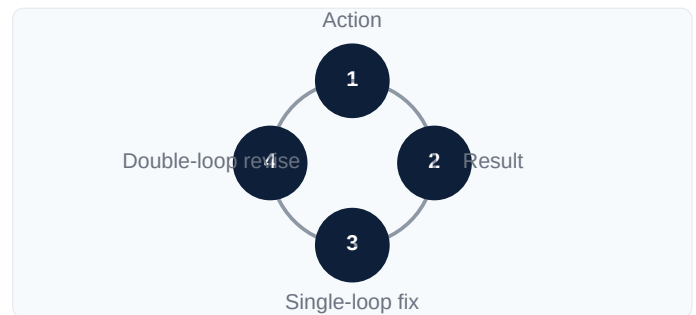


Figure 1. Single-loop fixes the error against existing rules; double-loop fixes the rules. Adapted from Argyris & Schön (1974).

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Where Lewin's (1947) action-research treated learning as discrete change-cycles, Argyris's contribution is the two-level distinction that makes meta-learning visible. Building on Bateson's (1972) levels-of-learning framework, double-loop learning gave organisational learning a teachable structural depth.

RELATED FRAMEWORKS IN THIS COURSE

Iceberg · 4 Levels · Slibrary 25

Surface vs Deep · Slibrary 25

Theory U · Slibrary 27

Sprint Retrospective · Slibrary 6

SOURCES (HARVARD REFERENCING STYLE)

Argyris, C. and Schön, D.A. (1974) *Theory in Practice: Increasing Professional Effectiveness*. San Francisco: Jossey-Bass.
 Argyris, C. and Schön, D.A. (1996) *Organizational Learning II: Theory, Method, and Practice*. Reading, MA: Addison-Wesley.
 Argyris, C. (2004) *Reasons and Rationalizations: The Limits to Organizational Knowledge*. Oxford: Oxford University Press.
 Bateson, G. (1972) *Steps to an Ecology of Mind*. New York: Ballantine.

12 Leverage Points

SLIBRARY 26 · SHEET 1 / 5

Meadows — twelve places to intervene, ranked by power

PROFESSIONAL DEFINITION

Donella Meadows's Twelve Leverage Points is a ranked taxonomy of places to intervene in a system, from least to most powerful: *numbers, buffers, stocks/flows, delays, balancing loops, reinforcing loops, information flows, rules, self-organisation, goals, paradigms, transcending paradigms*. The counter-intuitive insight is that the most-used interventions (numbers, parameters) are the weakest, while the most-powerful ones (paradigms, goals) are rarely attempted. The list inverts conventional managerial intuition (Meadows, 1999, 2008).

WHO DEVELOPED IT

Donella H. Meadows, environmental scientist and lead author of *The Limits to Growth* (1972), in her 1999 paper "Leverage Points: Places to Intervene in a System" published by the Sustainability Institute and elaborated posthumously in *Thinking in Systems* (2008). Built on Jay Forrester's (1961) system-dynamics tradition.

WHY IT WAS DEVELOPED

Practitioners intervening in systems used the lowest-leverage points (parameters, numbers) most often because they were easiest. Meadows argued — backed by decades of system-dynamics modelling — that the highest-leverage interventions were paradigms and goals, almost never attempted. The ranked list made the intuition-inversion explicit.

HOW IT IS USED

Diagnose where in the system intervention is being attempted. Map current intervention to a leverage point on the list. Test whether higher-leverage intervention (deeper in the system) is feasible. The harder the intervention, the more durable the change.

WHEN IT IS USED

Use in policy-intervention design, in transformation-programme architecture, in environmental-and-social-system design, and as the strategic frame for systems interventions across domains.

BY WHOM IT IS USED

Public-policy analysts, environmental-policy designers, transformation officers, senior leaders sponsoring deep change, and consultants advising on systems-level interventions.

FIGURE 1 · TWELVE POINTS (WEAKEST → STRONGEST)

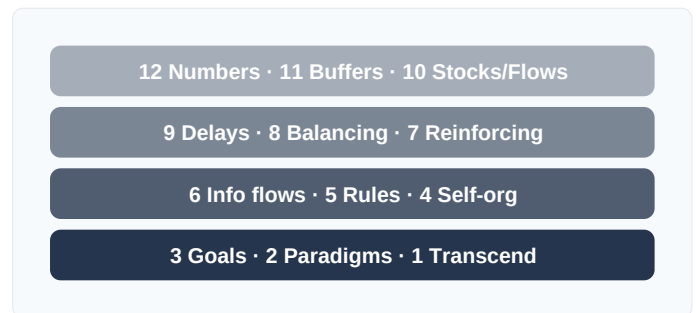


Figure 1. The list inverts conventional intuition; deepest interventions are most powerful but rarely attempted. Adapted from Meadows (1999).

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Where Forrester's (1961) system dynamics modelled systems mathematically and Senge's (1990) Iceberg framed intervention depth, Meadows's contribution is the ranked, prescriptive twelve-point list. Building on Bateson's (1972) levels of learning, it became the canonical systems-thinking intervention taxonomy.

RELATED FRAMEWORKS IN THIS COURSE

Parameters & Buffers · Slibrary 26

Feedback Loops · Slibrary 26

Goals of the System · Slibrary 26

Paradigm Shift · Slibrary 26

SOURCES (HARVARD REFERENCING STYLE)

Meadows, D.H. (1999) *Leverage Points: Places to Intervene in a System*. Hartland: Sustainability Institute.

Meadows, D.H. (2008) *Thinking in Systems: A Primer*. White River Junction: Chelsea Green.

Meadows, D.H. et al. (1972) *The Limits to Growth*. New York: Universe Books.

Forrester, J.W. (1961) *Industrial Dynamics*. Cambridge, MA: MIT Press.

Parameters & Buffers

SLIBRARY 26 · SHEET 2 / 5

Meadows points 12 & 11 — change numbers, change capacity

PROFESSIONAL DEFINITION

Parameters and Buffers are Meadows's lowest-leverage intervention points: *parameters* (numbers — taxes, subsidies, prices, standards) and *buffers* (stocks that absorb shock — inventory, savings, emergency capacity). Most policy and management intervention happens at these levels because they are tractable; durable systems change requires deeper interventions. The diagnostic value is recognising when the team is about to invest heavily in a low-leverage intervention (Meadows, 1999, 2008).

WHO DEVELOPED IT

Donella H. Meadows in her 1999 leverage-points paper, building on decades of system-dynamics modelling. The parameters-and-buffers diagnostic was elaborated in workshop practice by Meadows and her successors at the Donella Meadows Institute.

WHY IT WAS DEVELOPED

Most practitioners intervened at the parameter level (raise the price, increase the inventory) because it was tractable, but recurring problems persisted because deeper structural causes were untouched. Naming parameters and buffers as low-leverage points made the limitation explicit.

HOW IT IS USED

When intervention is proposed at the parameter or buffer level, ask whether deeper leverage points are accessible. Sometimes parameter changes are appropriate (genuine one-off rebalancing); often they are surface fixes for deeper problems.

WHEN IT IS USED

Use as a diagnostic before committing to parameter-level interventions, in policy design, in operations-improvement programmes, and in transformation-programme architecture decisions.

BY WHOM IT IS USED

Public-policy analysts, operations leaders, transformation officers, and senior leaders sponsoring change programmes that risk over-investing at the parameter level.

FIGURE 1 · LOWEST-LEVERAGE POINTS

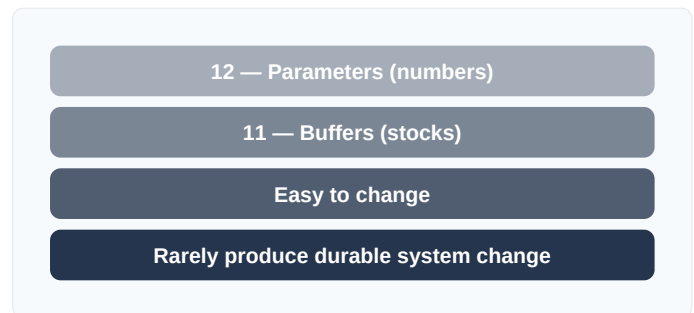


Figure 1. Parameter and buffer changes are the most-attempted and least-effective interventions.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Where conventional management intervened at parameters by default, Meadows's contribution is the explicit low-leverage classification that surfaces the limitation. Building on Forrester's (1961) system-dynamics models showing parameter-change limitations, it gives practitioners a diagnostic.

RELATED FRAMEWORKS IN THIS COURSE

12 Leverage Points · Slibrary 26

Feedback Loops · Slibrary 26

Goals of the System · Slibrary 26

Surface vs Deep · Slibrary 25

SOURCES (HARVARD REFERENCING STYLE)

Meadows, D.H. (1999) *Leverage Points: Places to Intervene in a System*. Hartland: Sustainability Institute.
 Meadows, D.H. (2008) *Thinking in Systems*. White River Junction: Chelsea Green.
 Forrester, J.W. (1961) *Industrial Dynamics*. Cambridge, MA: MIT Press.
 Serman, J.D. (2000) *Business Dynamics*. New York: McGraw-Hill.

Feedback Loops

SLIBRARY 26 · SHEET 3 / 5

Meadows points 8 & 7 — balancing dampens, reinforcing accelerates

PROFESSIONAL DEFINITION

Feedback Loops are Meadows's mid-leverage intervention points (balancing loops at #8, reinforcing loops at #7). *Balancing loops* dampen change toward a goal (thermostats, pricing equilibria); *reinforcing loops* amplify change in the same direction (compound interest, viral spread). Strengthening a balancing loop stabilises a system; weakening a runaway reinforcing loop prevents collapse. Loop intervention is more powerful than parameter change because it shifts system behaviour, not just numbers (Meadows, 1999, 2008; Sterman, 2000).

WHO DEVELOPED IT

Articulated by Donella Meadows in her 1999 leverage-points paper, building on Jay Forrester's (1961) system-dynamics tradition. Operationalised in management practice by John Sterman in *Business Dynamics* (2000) at MIT Sloan.

WHY IT WAS DEVELOPED

Practitioners intervened on system parameters without recognising the feedback loops generating system behaviour; many parameter changes were quickly absorbed by counteracting loops, defeating the intervention. Loop-level intervention addresses the dynamics directly, producing more durable change.

HOW IT IS USED

Map the system's feedback loops. Identify which balancing loops are too weak (system is unstable) or too strong (system is rigid). Identify which reinforcing loops are runaway (system is collapsing) or stalled (system isn't growing). Intervene at loop-strength rather than parameters.

WHEN IT IS USED

Use in supply-chain redesign, in policy intervention beyond parameter-change, in market-stability analysis, and in growth-engine design where reinforcing dynamics matter.

BY WHOM IT IS USED

Operations strategists, public-policy analysts, growth-strategy leaders, supply-chain designers, and consultants advising on systems-level interventions.

FIGURE 1 · TWO LOOP TYPES

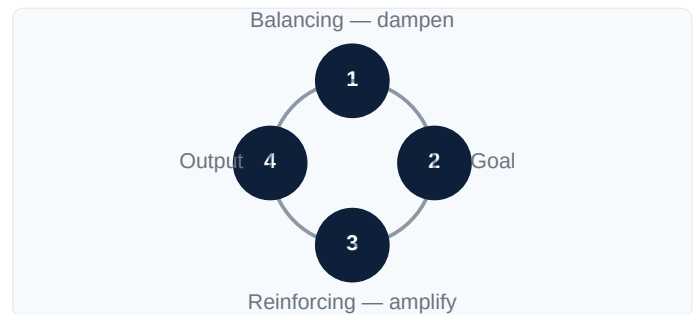


Figure 1. Balancing loops stabilise; reinforcing loops accelerate. Loop-strength change is more powerful than parameter change.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Where Forrester (1961) modelled feedback loops mathematically and Senge's (1990) Iceberg used loops descriptively, Meadows's contribution is the ranked-leverage classification that places loops above parameters but below goals and paradigms. Building on Wiener's (1948) cybernetic feedback theory, it gives practitioners loop-intervention as a deliberate move.

RELATED FRAMEWORKS IN THIS COURSE

12 Leverage Points · Slibrary 26

Parameters & Buffers · Slibrary 26

Systems Archetypes (8) · Slibrary 27

Iceberg · 4 Levels · Slibrary 25

SOURCES (HARVARD REFERENCING STYLE)

Meadows, D.H. (1999) *Leverage Points*. Hartland: Sustainability Institute.
 Sterman, J.D. (2000) *Business Dynamics*. New York: McGraw-Hill.
 Forrester, J.W. (1961) *Industrial Dynamics*. Cambridge, MA: MIT Press.
 Wiener, N. (1948) *Cybernetics*. Cambridge, MA: MIT Press.

Goals of the System

SLIBRARY 26 · SHEET 4 / 5

Meadows point 3 — change what the system is for, change everything else

PROFESSIONAL DEFINITION

Goals of the System is Meadows's third-highest leverage point: changing the goal that the entire system is designed to serve. A system optimised for one goal (shareholder returns) behaves fundamentally differently when re-purposed for another (employee wellbeing, ecological balance). Most parameters, rules and structures will adjust themselves once the goal changes. Goal change is rare because it threatens incumbent power, but it produces structural transformation that no number of parameter changes can match (Meadows, 1999, 2008).

WHO DEVELOPED IT

Donella Meadows in her 1999 leverage-points paper, ranked third behind only paradigms and transcendence. The goals-of-the-system concept builds on Ackoff's (1981) purposeful-systems theory and Stafford Beer's (1979) viable-system model.

WHY IT WAS DEVELOPED

Systems with rigid goals could not be reformed at parameter or rule levels — every change was eventually absorbed back to serve the original goal. Meadows argued that durable transformation required goal-level change, which most practitioners never attempted because of the political stakes.

HOW IT IS USED

Diagnose the system's actual operating goal (often different from stated goal). Identify whose interests the goal serves. Make goal-change the explicit strategic objective. Expect resistance from incumbents whose power depends on the existing goal.

WHEN IT IS USED

Use in deep transformation programmes, in mission-driven organisational redesign, in policy-reform contexts where parameter-change has repeatedly failed, and in cooperative or B-Corp formation decisions.

BY WHOM IT IS USED

Senior leaders sponsoring deep change, mission-driven founders, public-policy reformers, organisational-development consultants advising on transformation, and board members revising governance documents.

FIGURE 1 · GOAL LAYER

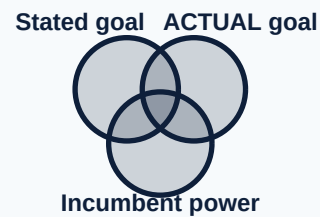


Figure 1. Operating goals often differ from stated goals; incumbent power depends on the actual goal.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Where Ackoff's (1981) purposeful-systems theory framed systems as goal-pursuing abstractly and Beer's (1979) viable-system model focused on operational viability, Meadows's contribution is the explicit goal-change-as-leverage-point ranking. Building on Forrester's (1961) system-dynamics, it gives practitioners goal-revision as a deliberate (though politically hard) move.

RELATED FRAMEWORKS IN THIS COURSE

12 Leverage Points · Slibrary 26

Paradigm Shift · Slibrary 26

Iceberg · 4 Levels · Slibrary 25

Theory U · Slibrary 27

SOURCES (HARVARD REFERENCING STYLE)

Meadows, D.H. (1999) *Leverage Points*. Hartland: Sustainability Institute.
 Meadows, D.H. (2008) *Thinking in Systems*. White River Junction: Chelsea Green.
 Ackoff, R.L. (1981) *Creating the Corporate Future*. New York: Wiley.
 Beer, S. (1979) *The Heart of Enterprise*. Chichester: Wiley.

Paradigm Shift

SLIBRARY 26 · SHEET 5 / 5

Meadows point 2 — change the mental model that underwrites the system

PROFESSIONAL DEFINITION

Paradigm Shift is Meadows's second-highest leverage point: changing the deepest mental models — beliefs about how the world works — that underwrite the system's goals, structures and parameters. Examples include the shift from Ptolemaic to Copernican cosmology, the shift from mainframe to personal computing, or from shareholder-primacy to stakeholder capitalism. Paradigm shifts are rare and politically explosive because they invalidate incumbent expertise and authority (Kuhn, 1962; Meadows, 1999).

WHO DEVELOPED IT

Thomas Kuhn introduced the paradigm-shift concept in *The Structure of Scientific Revolutions* (1962) for scientific contexts; Donella Meadows applied it to systems-intervention as the second-highest leverage point in her 1999 paper, ranked above goals.

WHY IT WAS DEVELOPED

Paradigm shifts had been treated as historical phenomena that just happened. Meadows argued they could be deliberately engineered as system interventions — though the engineering was orders of magnitude harder than parameter, rule, or goal change. Naming the paradigm level made it discussable as an intervention target.

HOW IT IS USED

Identify the dominant paradigm. Surface its assumptions. Demonstrate the paradigm's failures empirically. Articulate an alternative paradigm coherently. Find allies whose interests are served by the new paradigm. Make the shift visible through repeated paradigm-affirming actions and stories.

WHEN IT IS USED

Use in deep social-and-economic transformation contexts, in industries facing platform shifts, in mission-driven movement-building, and as the long-term strategic frame for systems-change leadership.

BY WHOM IT IS USED

Movement leaders, mission-driven founders, public-intellectual leaders, transformation officers in industries facing fundamental shifts, and academic and policy leaders.

FIGURE 1 · PARADIGM UNDERWRITES ALL

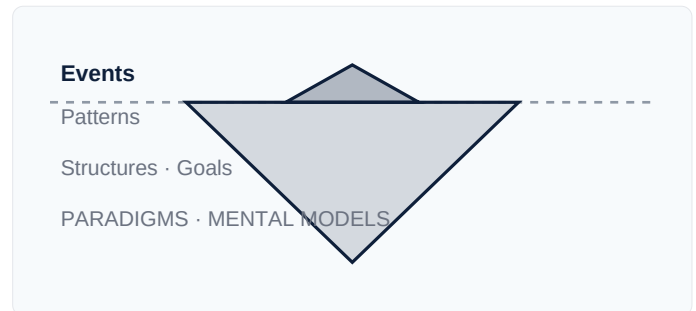


Figure 1. Paradigms underwrite goals and structures; changing them changes everything beneath.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Where Kuhn's (1962) paradigm-shift concept applied to scientific revolutions, Meadows's contribution is its application to general systems-intervention as a deliberate (if difficult) leverage move. Building on Lakatos's (1970) research-programme tradition, it gives systems-change leaders the deepest intervention frame short of transcendence.

RELATED FRAMEWORKS IN THIS COURSE

12 Leverage Points · Slibrary 26

Goals of the System · Slibrary 26

Theory U · Slibrary 27

Iceberg · 4 Levels · Slibrary 25

SOURCES (HARVARD REFERENCING STYLE)

Kuhn, T.S. (1962) *The Structure of Scientific Revolutions*. Chicago: University of Chicago Press.

Meadows, D.H. (1999) *Leverage Points*. Hartland: Sustainability Institute.

Meadows, D.H. (2008) *Thinking in Systems*. White River Junction: Chelsea Green.

Lakatos, I. (1970) 'Falsification and the Methodology of Scientific Research Programmes', in *Criticism and the Growth of Knowledge*. Cambridge: Cambridge University Press.

Systems Archetypes (8)

SLIBRARY 27 · SHEET 1 / 5

Eight recurring system patterns — recognise them, change behaviour

PROFESSIONAL DEFINITION

Systems Archetypes are eight recurring patterns of system behaviour first catalogued by Senge (1990): *Limits to Growth*, *Shifting the Burden*, *Eroding Goals*, *Escalation*, *Success to the Successful*, *Tragedy of the Commons*, *Fixes that Fail*, *Growth and Underinvestment*. Each archetype describes a structural pattern — combinations of feedback loops and delays — that produces predictable problems across domains. Recognising the archetype gives practitioners the leverage point that resolves it (Senge, 1990; Kim and Anderson, 1998).

WHO DEVELOPED IT

Peter Senge codified eight archetypes in *The Fifth Discipline* (1990); Daniel Kim and Virginia Anderson elaborated and visualised them in *Systems Archetype Basics* (1998). Roots in Forrester's (1961) system-dynamics tradition and the Pegasus Communications systems-thinking community.

WHY IT WAS DEVELOPED

Practitioners encountered the same structural problems repeatedly across domains without recognising the pattern. Senge's archetypes named the patterns explicitly, giving practitioners pattern-recognition shortcuts and pre-validated leverage points for each archetype.

HOW IT IS USED

Diagnose system behaviour against the eight archetypes. Match observed pattern to archetype. Apply the archetype-specific leverage point: e.g., for *Shifting the Burden*, strengthen the fundamental solution; for *Tragedy of the Commons*, regulate the shared resource.

WHEN IT IS USED

Use in systems-thinking workshops, in chronic-problem diagnosis, in policy analysis, in organisational-development interventions, and as a teaching frame for systems-thinking practice.

BY WHOM IT IS USED

Systems-thinking practitioners, organisational-development consultants, public-policy analysts, transformation officers, and senior leaders facing recurring structural problems.

FIGURE 1 · EIGHT ARCHETYPES



Figure 1. Eight named patterns; each carries a pre-validated leverage point. Adapted from Senge (1990).

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Where Forrester's (1961) system dynamics required mathematical modelling, Senge's contribution is the named-pattern catalogue that makes systems thinking accessible without simulation. Building on Christopher Alexander's (1977) pattern-language tradition, it gave systems thinking a teachable typology.

RELATED FRAMEWORKS IN THIS COURSE

Shifting the Burden · Slibrary 27

Tragedy of the Commons · Slibrary 27

Feedback Loops · Slibrary 26

Iceberg · 4 Levels · Slibrary 25

SOURCES (HARVARD REFERENCING STYLE)

Senge, P.M. (1990) *The Fifth Discipline*. New York: Doubleday.
 Kim, D.H. and Anderson, V. (1998) *Systems Archetype Basics*. Waltham: Pegasus Communications.
 Serman, J.D. (2000) *Business Dynamics*. New York: McGraw-Hill.
 Alexander, C. (1977) *A Pattern Language*. New York: Oxford University Press.

Shifting the Burden

SLIBRARY 27 · SHEET 2 / 5

Symptomatic fix preferred → fundamental solution atrophies

PROFESSIONAL DEFINITION

Shifting the Burden is the systems archetype in which a symptomatic fix is repeatedly preferred over a fundamental solution because the symptomatic fix is faster and easier. Over time, capacity for the fundamental solution atrophies, and dependence on the symptomatic fix grows — until the fundamental solution is no longer accessible. Examples include painkiller addiction substituting for treating root cause, outsourcing replacing internal capability, and emergency funding replacing structural reform (Senge, 1990).

WHO DEVELOPED IT

Peter Senge in *The Fifth Discipline* (1990) as one of the eight original systems archetypes. Extends Argyris and Schön's (1974) theories-in-use research and Sterman's (2000) systems-dynamics modelling of substance-dependence dynamics.

WHY IT WAS DEVELOPED

Organisations and societies repeatedly chose symptomatic fixes over fundamental solutions and were puzzled when the fundamental solution became unavailable. The archetype made the dynamic explicit — preference + atrophy = irreversibility — and prescribed prevention through deliberate fundamental-solution investment.

HOW IT IS USED

Diagnose whether a symptomatic fix is being preferred. Map the atrophy of fundamental-solution capacity. Strengthen the fundamental solution explicitly even though it's slower. Set up structural commitment that prevents over-reliance on the symptomatic fix.

WHEN IT IS USED

Use in addiction-treatment design, in outsourcing decisions, in chronic-policy intervention, in organisational-capability development, and whenever a quick fix is being chosen repeatedly.

BY WHOM IT IS USED

Public-health policy analysts, organisational-development consultants, change-management leaders, capability-development directors, and senior leaders sponsoring transformation programmes.

FIGURE 1 · BURDEN SHIFT

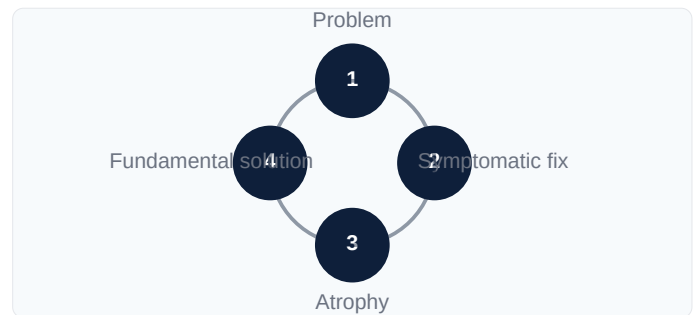


Figure 1. Symptomatic fix preference + fundamental atrophy = irreversibility. The leverage is fundamental-investment.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Where Argyris and Schön's (1974) theories-in-use research described the cognitive bias toward easier fixes, Senge's contribution is the named-archetype with explicit atrophy mechanism. Building on Forrester's (1961) system dynamics, it became a canonical archetype with prescribed leverage.

RELATED FRAMEWORKS IN THIS COURSE

Systems Archetypes (8) · Slibrary 27

Tragedy of the Commons · Slibrary 27

Surface vs Deep · Slibrary 25

Double-Loop Learning · Slibrary 25

SOURCES (HARVARD REFERENCING STYLE)

- Senge, P.M. (1990) *The Fifth Discipline*. New York: Doubleday.
 Kim, D.H. and Anderson, V. (1998) *Systems Archetype Basics*. Waltham: Pegasus Communications.
 Sterman, J.D. (2000) *Business Dynamics*. New York: McGraw-Hill.
 Argyris, C. and Schön, D.A. (1974) *Theory in Practice*. San Francisco: Jossey-Bass.

Tragedy of the Commons

SLIBRARY 27 · SHEET 3 / 5

Shared resource × individual incentive → collective collapse

PROFESSIONAL DEFINITION

The Tragedy of the Commons is the systems archetype — and a foundational concept in environmental and resource economics — describing how individually rational use of a shared resource leads to collective overuse and collapse. Each user gains the full benefit of one more unit of use while bearing only a fraction of the cumulative cost. Without governance structures (regulation, privatisation, community management), shared resources are systematically over-exploited (Hardin, 1968; Ostrom, 1990; Senge, 1990).

WHO DEVELOPED IT

Garrett Hardin in his 1968 *Science* paper "The Tragedy of the Commons"; Elinor Ostrom in *Governing the Commons* (1990) demonstrated empirically that community-managed commons could resolve the tragedy without privatisation, winning the Nobel Prize in Economics (2009). Senge (1990) named it as a systems archetype.

WHY IT WAS DEVELOPED

Environmental and resource problems were repeatedly attributed to bad actors when the structural cause was the commons-incentive pattern itself. Hardin's contribution was to name the structural inevitability under unmanaged conditions; Ostrom's contribution was to demonstrate that community governance could resolve it.

HOW IT IS USED

Diagnose whether a problem fits the commons pattern: shared resource, individual incentive to over-use, weak collective governance. Apply Ostrom's (1990) eight design principles for governing commons: clear boundaries, congruent rules, collective choice, monitoring, graduated sanctions, conflict resolution, recognition, nested enterprises.

WHEN IT IS USED

Use in environmental-policy design, in shared-infrastructure management, in platform-governance design, in shared-organisational-resource allocation, and in shared-data and IP-pool design.

BY WHOM IT IS USED

Environmental-policy analysts, public-policy designers, platform-governance leaders, organisational-development consultants advising on shared-resource governance, and academic researchers in commons studies.

FIGURE 1 · COMMONS DYNAMIC

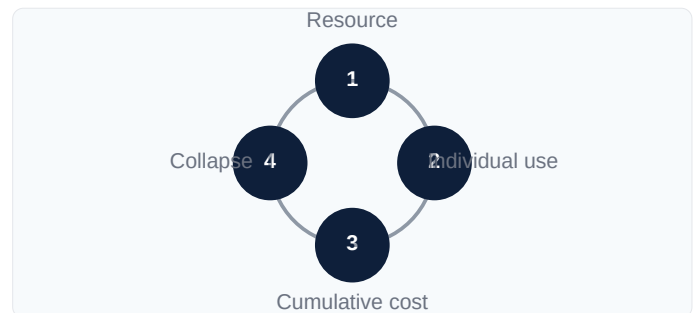


Figure 1. Individual benefit > individual cost share → systematic over-use → collapse without governance.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Where Hardin's (1968) original paper argued for either privatisation or strong state regulation, Ostrom's contribution is the empirical demonstration that community-governance could work. Building on Schelling's (1978) micromotives-and-macrobehavior tradition, it gave commons-management a third path beyond market and state.

RELATED FRAMEWORKS IN THIS COURSE

Systems Archetypes (8) · Slibrary 27

Shifting the Burden · Slibrary 27

Goals of the System · Slibrary 26

Feedback Loops · Slibrary 26

SOURCES (HARVARD REFERENCING STYLE)

Hardin, G. (1968) 'The Tragedy of the Commons', *Science*, 162(3859), pp. 1243–1248.

Ostrom, E. (1990) *Governing the Commons: The Evolution of Institutions for Collective Action*. Cambridge: Cambridge University Press.

Senge, P.M. (1990) *The Fifth Discipline*. New York: Doubleday.

Schelling, T.C. (1978) *Micromotives and Macrobehavior*. New York: W.W. Norton.

Theory U

SLIBRARY 27 · SHEET 4 / 5

Scharmer — Sense · Presence · Realise — leading from the emerging future

PROFESSIONAL DEFINITION

Otto Scharmer's Theory U is a leadership framework for systems-change that traces a U-shaped journey through three movements: *Sensing* (perceiving the current reality with fresh eyes), *Presencing* (connecting to the deepest source of intention from which the emerging future can be sensed), and *Realising* (acting from that source to bring the new into being). Theory U explicitly addresses the paradigm-and-mental-model leverage points where most systems-change leaders never go — making the inner shift the precondition for outer change (Scharmer, 2009, 2018).

WHO DEVELOPED IT

C. Otto Scharmer at MIT Sloan in *Theory U: Leading from the Future as It Emerges* (2009) and the *Essentials of Theory U* (2018), built on the Presencing Institute's twenty years of practice with multinational corporations, governments and NGOs. Influences include Jung (collective unconscious), Buddhist contemplative practice, and action-research traditions.

WHY IT WAS DEVELOPED

Conventional change-management addressed visible structures and behaviours but failed at deep transformation because the leaders' own mental models remained unchanged. Theory U made the leader's inner shift the explicit precondition for systems change — making transformation intervention as much an inner journey as an outer one.

HOW IT IS USED

Walk the U: *Co-initiate* (uncover common intent), *Co-sense* (observe with fresh eyes), *Co-presence* (connect to source), *Co-create* (prototype new), *Co-evolve* (embody in ecosystems). Each movement requires specific practices that defeat habitual perception and judgement.

WHEN IT IS USED

Use in deep transformation programmes, in mission-driven movement-building, in inter-sector collaboration on wicked problems, and in leadership-development contexts where outer change requires inner work.

BY WHOM IT IS USED

Senior leaders sponsoring deep transformation, social entrepreneurs, mission-driven movement leaders, executive coaches working with systems-change leaders, and academic researchers in leadership development.

FIGURE 1 · U-SHAPED JOURNEY

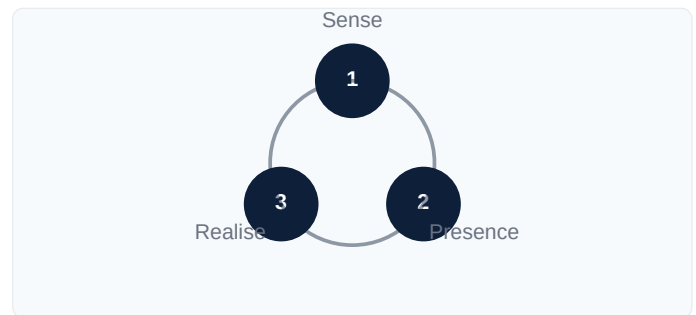


Figure 1. The U descends to source before ascending into action; the inner shift precedes the outer. Adapted from Scharmer (2009).

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Where Senge's (1990) *Fifth Discipline* framed organisational learning at structural level and Argyris's (1974) double-loop learning addressed cognitive depth, Scharmer's contribution is the explicit inner-leader-shift dimension. Building on Jung's (1959) archetype theory and contemplative-practice traditions, it gives systems leadership its deepest dimension.

RELATED FRAMEWORKS IN THIS COURSE

Presencing · Slibrary 27

Paradigm Shift · Slibrary 26

Double-Loop Learning · Slibrary 25

Goals of the System · Slibrary 26

SOURCES (HARVARD REFERENCING STYLE)

Scharmer, C.O. (2009) *Theory U: Leading from the Future as It Emerges*. San Francisco: Berrett-Koehler.

Scharmer, C.O. (2018) *The Essentials of Theory U*. San Francisco: Berrett-Koehler.

Senge, P., Scharmer, C.O., Jaworski, J. and Flowers, B.S. (2004) *Presence: Human Purpose and the Field of the Future*. Cambridge, MA: SoL.

Jung, C.G. (1959) *The Archetypes and the Collective Unconscious*. Princeton: Princeton University Press.

Presencing

SLIBRARY 27 · SHEET 5 / 5

Scharmer — letting go of past, letting come of future

PROFESSIONAL DEFINITION

Presencing is the bottom of Theory U — Scharmer's core construct combining "presence" and "sensing" — the moment in which the leader lets go of past assumptions and lets come the highest future possibility. Presencing is not technique but practice: a discipline of attention that defeats habitual perception, ego-driven judgement, and reactive emotion long enough to perceive what the emerging future is asking. The state is rare, real, and trainable through specific practices (Scharmer, 2009; Senge et al., 2004).

WHO DEVELOPED IT

C. Otto Scharmer with Peter Senge, Joseph Jaworski and Betty Sue Flowers in *Presence* (2004) and elaborated by Scharmer in *Theory U* (2009). Influences include Buddhist mindfulness traditions, Jungian active imagination, and Goethean phenomenology.

WHY IT WAS DEVELOPED

Leaders attempting deep transformation typically operated from past patterns even while claiming to lead toward the future. Presencing named the specific psychological state in which past-pattern-driven action is suspended long enough to perceive emerging-future requirements directly.

HOW IT IS USED

Practise the disciplines of Theory U: open mind (suspend judgement), open heart (suspend cynicism), open will (suspend fear). Engage practices: mindful walking, deep listening, journaling. Presencing arrives, not on demand, but through repeated discipline.

WHEN IT IS USED

Use in deep transformation work, in inter-sector collaboration on wicked problems, in personal leadership development, and as the contemplative practice underlying Theory U's outer-change methodology.

BY WHOM IT IS USED

Senior leaders sponsoring deep transformation, social entrepreneurs, mission-driven movement leaders, executive coaches, and academic researchers in leadership-development and contemplative practice.

FIGURE 1 · THREE OPENINGS

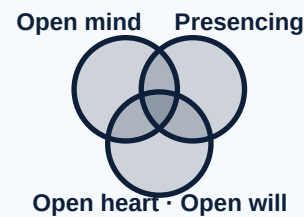


Figure 1. The three openings defeat habitual perception long enough for emerging-future sensing. Adapted from Scharmer (2009).

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Where mindfulness traditions (Kabat-Zinn, 1990) brought contemplative practice into clinical and corporate contexts and Goethean phenomenology (Goethe, 1810) emphasised receptive perception, Scharmer's contribution is the explicit linkage to systems-change leadership. Building on Jung's (1959) active-imagination tradition, it gives transformation its inner-practice precondition.

RELATED FRAMEWORKS IN THIS COURSE

Theory U · Slibrary 27

Paradigm Shift · Slibrary 26

Double-Loop Learning · Slibrary 25

Goals of the System · Slibrary 26

SOURCES (HARVARD REFERENCING STYLE)

Scharmer, C.O. (2009) *Theory U: Leading from the Future as It Emerges*. San Francisco: Berrett-Koehler.

Senge, P., Scharmer, C.O., Jaworski, J. and Flowers, B.S. (2004) *Presence: Human Purpose and the Field of the Future*. Cambridge, MA: SoL.

Kabat-Zinn, J. (1990) *Full Catastrophe Living*. New York: Delacorte.

Goethe, J.W. von (1810) *Theory of Colours*. Translated by C.L. Eastlake. London: Murray, 1840.